Website Performance Analysis and User Engagement Insights

---

1. Introduction

This project focuses on analyzing website performance and user engagement using real-world web analytics data. The goal is to identify performance issues, understand user behavior, and provide data-driven recommendations to improve overall website efficiency.

The dataset includes key metrics such as sessions, users, engagement rate, and event counts collected over time across different channels.

---

2. Problem Statement

Websites often face challenges like:

- Slow page load times
- Low user engagement
- High bounce rates
- Inconsistent performance across devices and browsers

This project aims to analyze these issues and identify factors affecting website performance and user interaction.

---

3. Objectives

- Analyze website traffic and engagement patterns
- Identify high-performing and low-performing segments
- Understand impact of devices, browsers, and time
- Detect anomalies (outliers in performance)
- Provide actionable recommendations

---

4. Dataset Description

- **Source:** AmanXai
- **Year:** 2024
- **Total Records:** 3182 rows
- **Total Features:** 10 columns

Key Features:

- Users
- Sessions
- Engaged Sessions
- Engagement Rate
- Events per Session
- Event Count
- Average Engagement Time
- Date & Hour
- Channel Group

---

5. Data Preprocessing

The dataset was cleaned and prepared using the following steps:

- Removed missing/null values
- Checked and eliminated duplicate records
- Converted date-time formats
- Ensured consistency in categorical data
- Verified numerical data ranges

This step ensured accurate and reliable analysis.

---

6. Exploratory Data Analysis (EDA)

User & Traffic Analysis

- Traffic varies significantly across different channels
- Some channels generate high sessions but low engagement

Engagement Analysis

- Engagement rate is not directly proportional to sessions
- Some users perform many actions but spend less time

Time-Based Analysis

- Peak usage occurs at specific hours
- Low activity observed during off-hours

Device & Performance Insights

- Mobile devices show higher page load times
- Desktop users experience better performance

Browser Analysis

- Performance differs across browsers
- Some browsers (e.g., Chrome) are faster

---

7. Visualization Techniques Used

- **Histograms** → Distribution of engagement metrics
- **Boxplots** → Detection of outliers
- **Bar Charts** → Channel-wise comparison
- **Time Series Plots** → Traffic trends

---

8. Key Findings

1. **Mobile Performance Issue**
    - Higher load times compared to desktop
    - Needs optimization
2. **Browser Variability**
    - Some browsers perform better than others
3. **Outliers Detected**
    - Certain pages have extremely high load times
4. **Engagement Gap**
    - High sessions ≠ High engagement
5. **Traffic Patterns**
    - Clear peak and low usage periods

---

9. Business Insights

- High traffic does not guarantee user engagement
- Poor performance directly impacts user experience
- Mobile users are at a disadvantage
- Optimization can significantly improve conversions

---

10. Recommendations

Performance Optimization

- Reduce page load time (especially mobile)
- Optimize images and scripts

Mobile Optimization

- Use responsive design
- Improve mobile UI/UX

Browser Compatibility

- Test across multiple browsers
- Fix inconsistencies

User Engagement

- Improve content quality
- Reduce unnecessary clicks

Monitoring

- Track performance regularly
- Set alerts for abnormal spikes

---

11. Conclusion

The analysis clearly shows that website performance and user engagement are influenced by multiple factors such as device type, browser, and traffic patterns. Improving mobile performance, optimizing slow pages, and enhancing user experience can significantly boost engagement and overall website success.

---

12. Future Scope

- Implement machine learning for prediction
- Real-time analytics dashboard
- A/B testing for performance improvements
- User segmentation for personalized experience

✅ **Final Summary**

This project successfully:

- Identified performance issues
- Analyzed user behavior
- Provided practical business recommendations

It demonstrates how data analytics can directly improve website performance and user satisfaction.